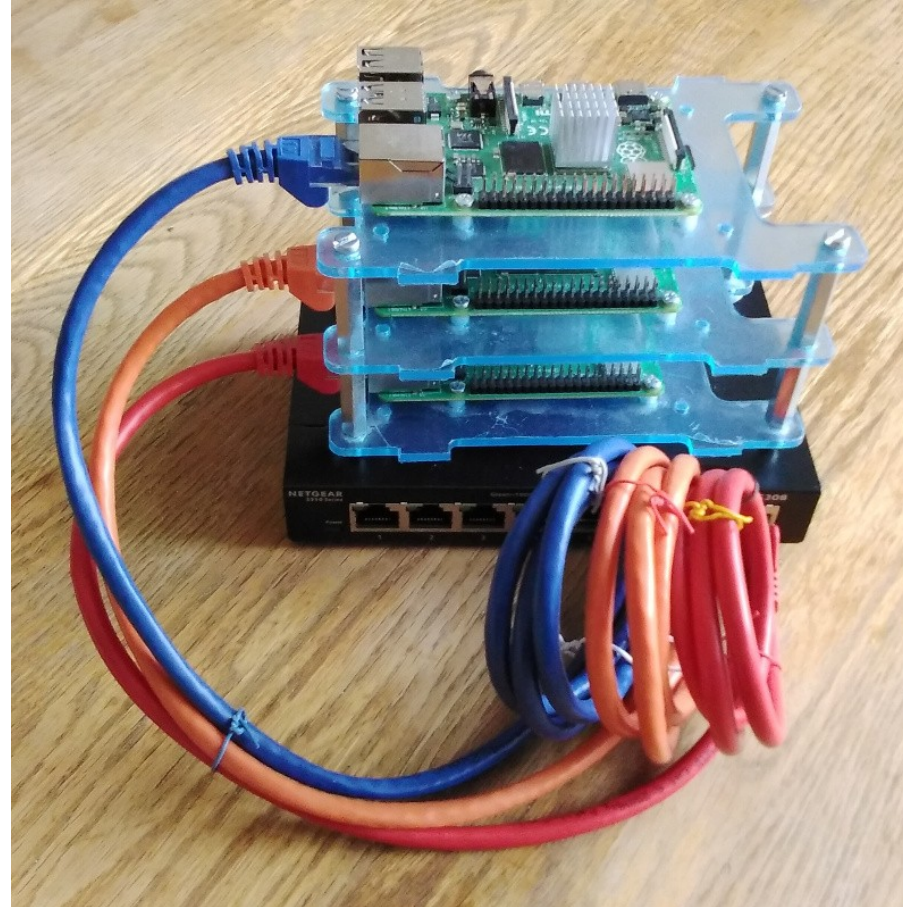


A Minimalist Network on Chip on an FPGA

Richard Miller `<r.miller@acm.org>`

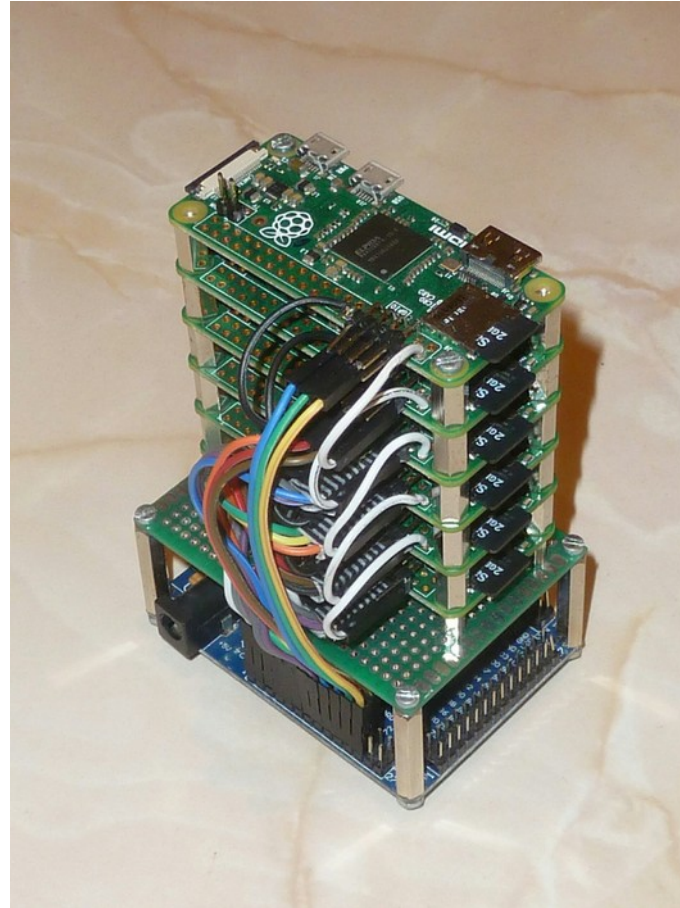
Open Source SG

The motivating problem: Ethernet + TCP/IP is unwieldy for linking a Raspberry Pi cluster



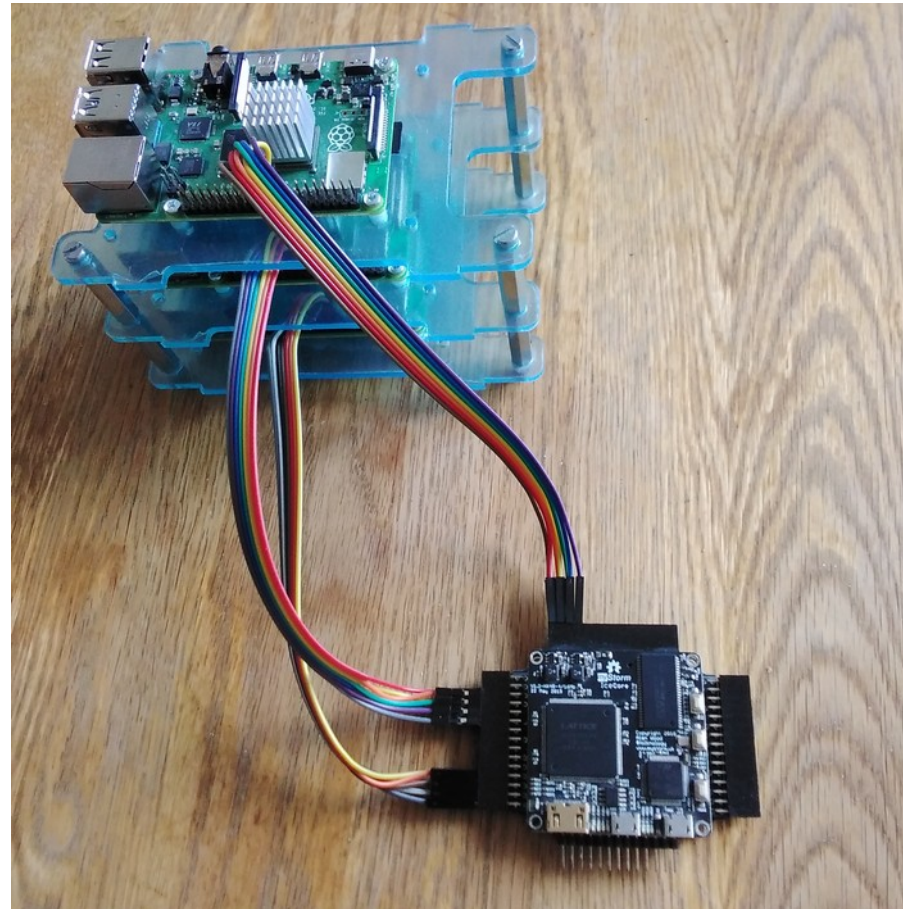
An alternative: create a simple network on an FPGA

First prototype: Raspberry Pi Zero cluster, Altera Cyclone 2 FPGA, connected via UART

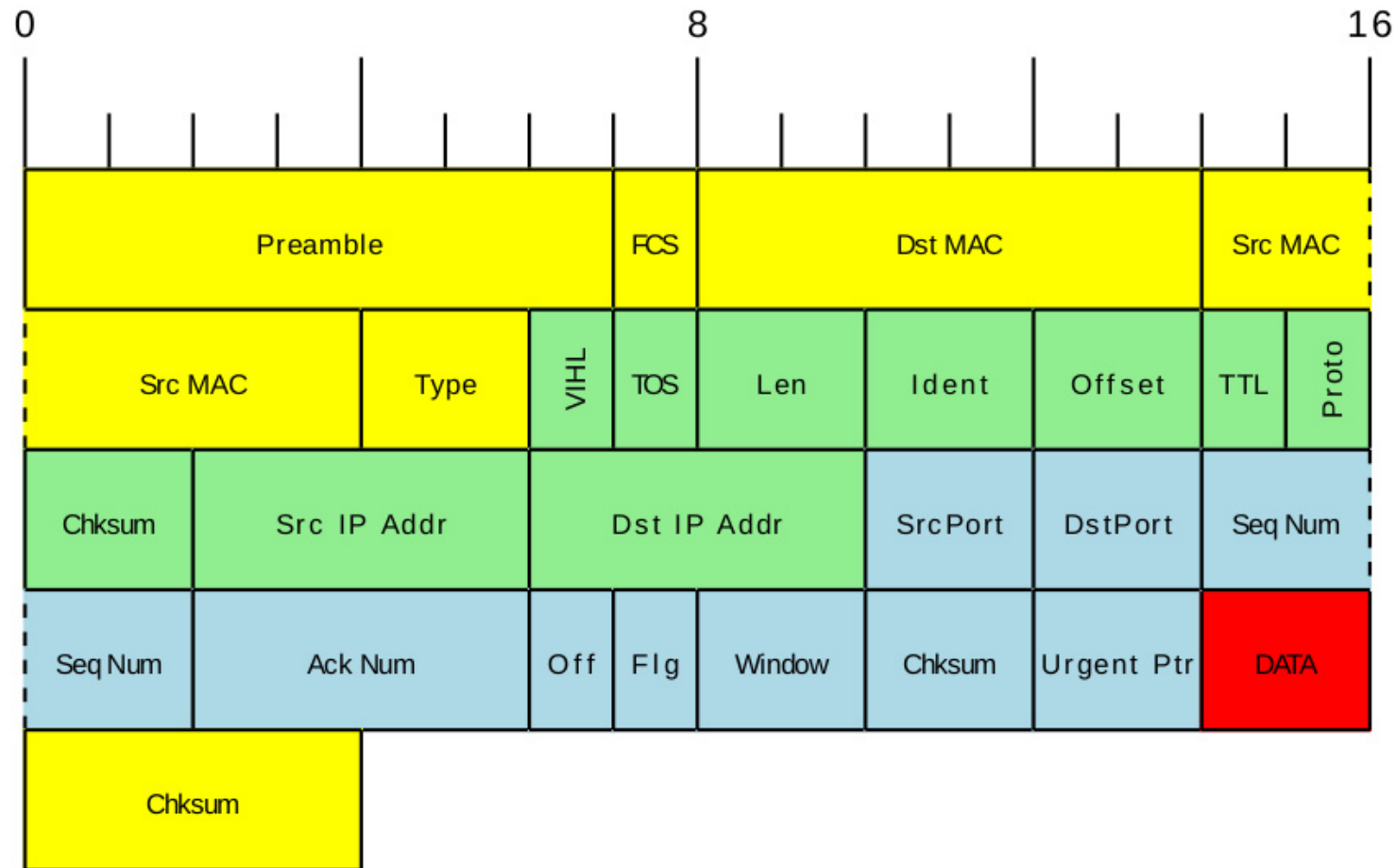


Current development platform: Raspberry Pi 4 cluster, BlackIce MX board with an ICE40 FPGA, connected via SPI

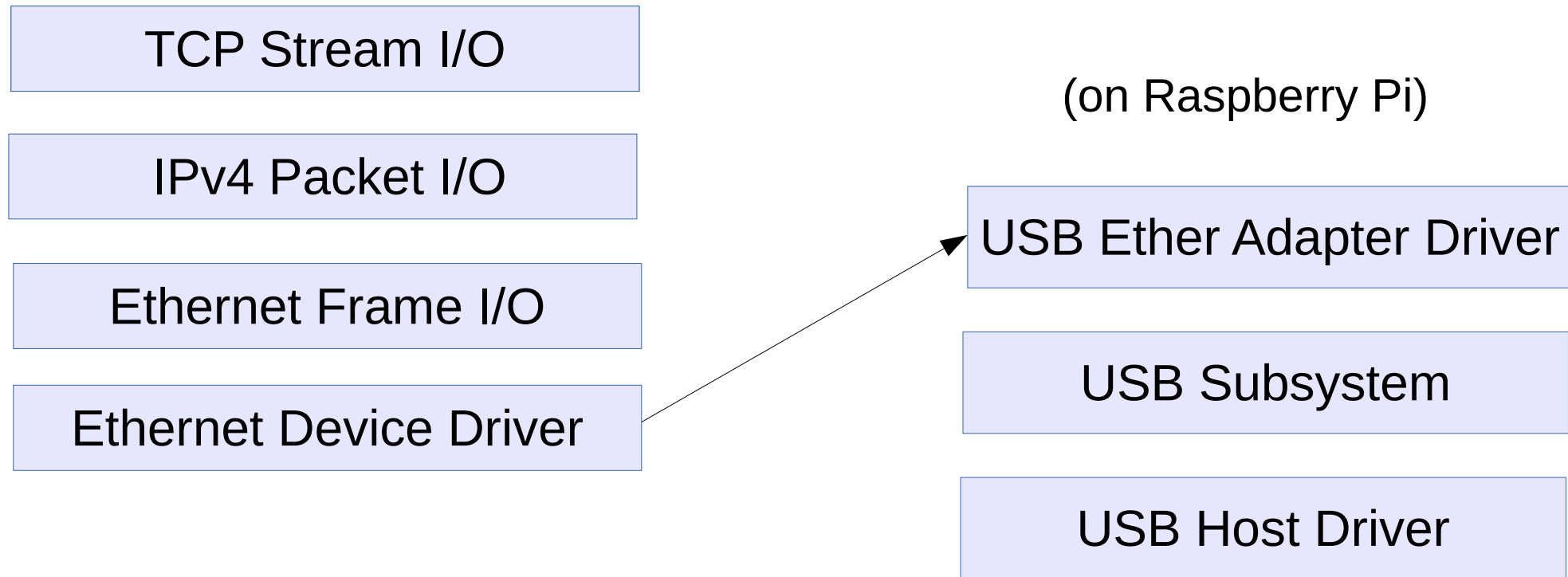
Advantage: open source FPGA toolchain (yosys + nextpnr)



Protocol layers $\Rightarrow$ extra headers $\Rightarrow$ more data on the wire

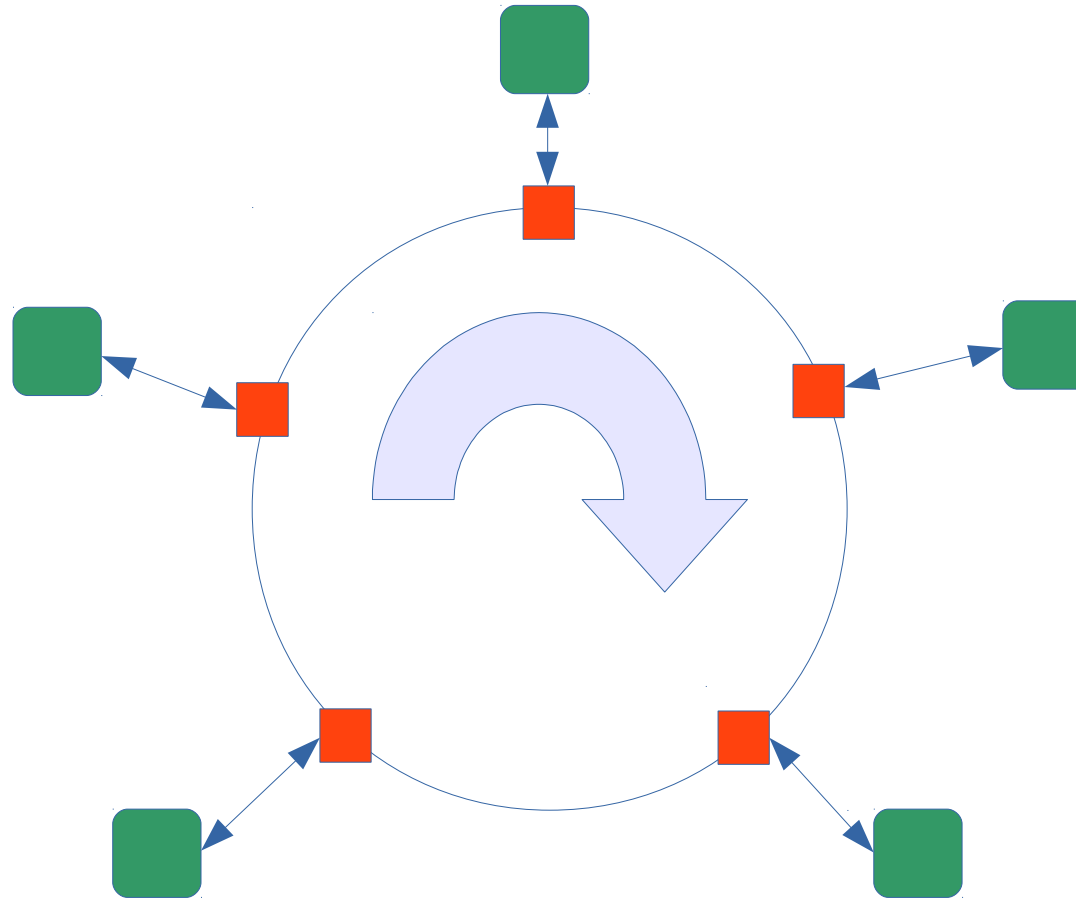


Protocol layers $\Rightarrow$ OS software layers $\Rightarrow$ extra latency

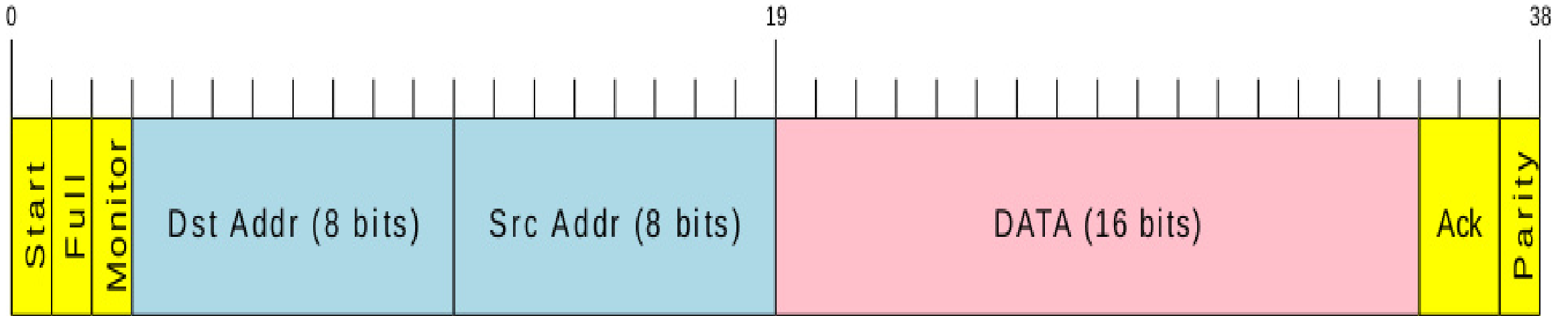


It doesn't have to be so complicated

The Cambridge digital communication ring, Wilkes & Wheeler 1979



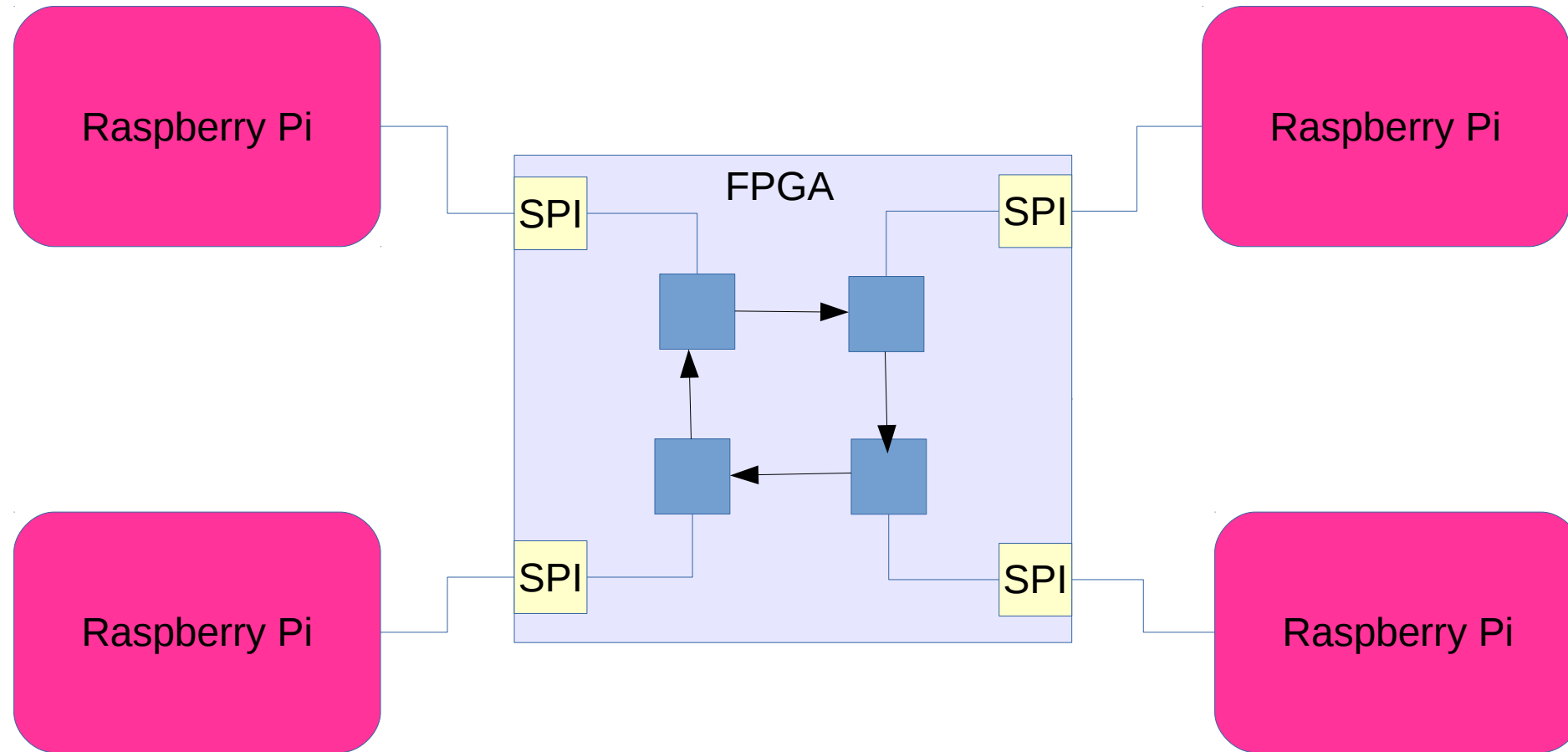
Cambridge ring packet (fixed length)



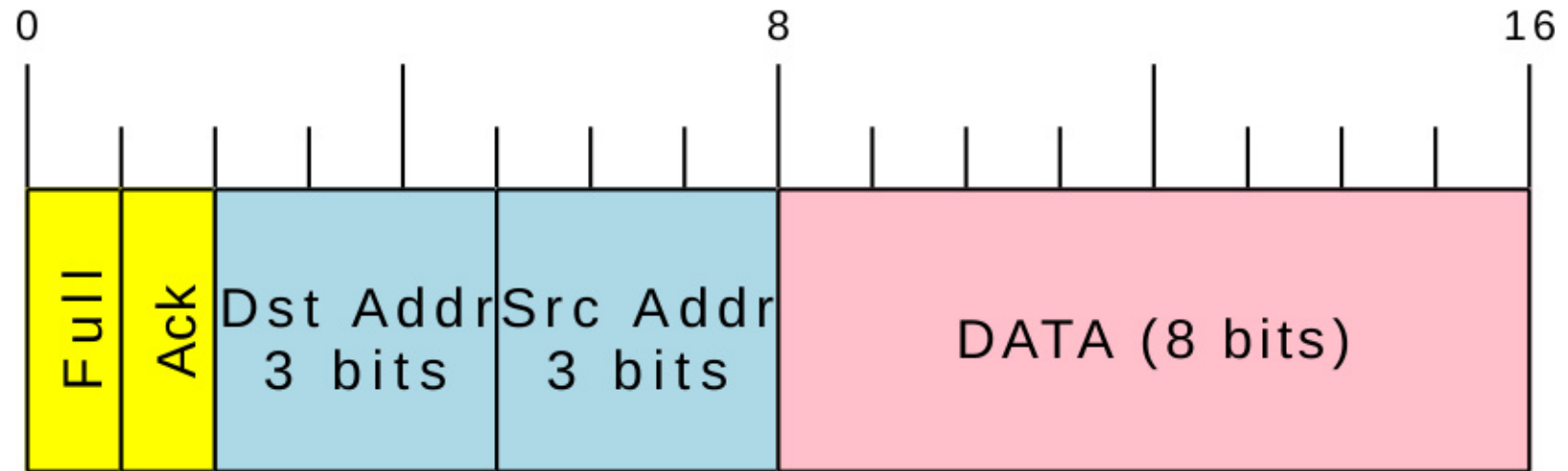
A later version extended the packet size to 32 bytes of data and 16-bit destination and source addresses.

The Cambridge Fast Ring Networking System, Hopper & Needham 1988

Implementing a simple ring network on an FPGA



For a small cluster, the packet can be even simpler



The client OS implementation (on Plan 9) is simple too

Network MUX

SPI Driver

```
/* receive from network node 3 and echo back */  
  
fd = open("/dev/spinet/3", ORDWR);  
while ((n = read(fd, buf, sizeof buf)) > 0)  
    write(fd, buf, n);
```

The ring node logic can be repurposed as a NoC for a SoC

